No External Dimensional Scale at the Level of Admissibility

Jérôme Beau
Independent Researcher
jerome.beau@cosmochrony.org

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Abstract

The admissibility structure of Foundation M [4] is built on a single primitive dimensional scale c_χ entering the Born–Infeld bound (A3–A4). We establish a structural rigidity result (Level 1): any deformation of the admissibility constraints that introduces a dimensional parameter independent of c_χ breaks at least one of BFS-consistent spectral scaling, structural non-injectivity (A2/ENI), or bounded admissibility closure (A3–A4). As an immediate corollary, dynamical fields carrying an independent dimensional coupling cannot be consistently coupled to the projection Π; this provides the structural underpinning for the exclusion of Reissner–Nordström and Schwarzschild–de Sitter geometries invoked in the companion paper Q6b. A stronger Level 2 statement, asserting full functional determination of all emergent scales, is beyond the scope of the present note and is identified as an open problem.

Keywords. admissibility; dimensional analysis; scale rigidity; non-injective projection; Born–Infeld constraint; spectral scaling; Mosco convergence; structural rigidity; emergent physics; admissibility cascade; effective scale

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1 Introduction and Motivation

Guiding principle

The admissibility structure of the Cosmochrony programme [6] is closed under reparameterisations of the primitive scale c_χ, but not under the introduction of independent dimensional parameters. The present note makes this principle precise.

Motivation

The companion paper Q6b establishes that the effective spacetime geometry derived from the admissibility operator $\mathcal{L}_{\text{eff}}$ selects the Schwarzschild metric as the unique exterior stationary solution compatible with spherical symmetry and bounded admissibility. The exclusion of the Reissner–Nordström and Schwarzschild–de Sitter deformations rests on the claim that no dimensional parameter independent of c_χ enters the admissibility structure: a charge Q with dimensions [charge] or a cosmological constant Λ with dimensions $[\text{length}]^{-2}$ would each introduce such a parameter. The present note supplies the structural backing for this claim.

Scope and limitations

The result proved here (Level 1) establishes that no independent dimensional parameter can be introduced into the admissibility constraints without breaking structural consistency. It does not establish Level 2: that every emergent scale is uniquely determined as a functional of c_χ and the cascade, without free parameters. The distinction is discussed in Section 5.

2 Framework and Definitions

We recall the admissibility axioms of Foundation M [4] in the form needed for the present argument. Throughout, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is the discrete Heisenberg group at prime q, $\{S_n\}$ is the BFS shell stratification, λ_n is the spectral quantity associated to the n-th admissibility window, and A_n is the admissibility amplitude.

(A2) The projection $\Pi\colon \mathcal{C} \to \mathcal{O}$ is non-injective; distinct underlying configurations may share the same observable image.

(A3) The Born–Infeld admissibility bound holds: $A_n \leq c_\chi/\sqrt{\lambda_n}$, where c_χ is the unique primitive dimensional scale of the admissibility structure.

(A4) The admissibility closure is bounded: the projected spectral capacity $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ on conjugate Weil blocks satisfies the power-law constraint established in [1].

Definition 2.1 (Independent dimensional parameter)**.** *A parameter λ is called an* independent dimensional parameter *(with respect to the admissibility structure) if the dimensions of λ are not expressible as a rational power of the dimensions of c_χ. Equivalently, $[\lambda] \neq [c_\chi]^k$ for any $k \in \mathbb{Q}$.*

A dimensionless *parameter (i.e., $[\lambda] = 1$) is not an independent dimensional parameter in this sense. In particular, any parameter entering the admissibility structure only through a dimensionless ratio with c_χ (i.e., through λ/c_χ^k for some $k \in \mathbb{Q}$) is not independent in the sense of this definition, and reduces to a rescaling of c_χ. Finite resolution parameters such as $q \in \mathbb{Z}_{>0}$ are dimensionless and disappear in the $q \to \infty$ continuum limit; they are excluded from the scope of the present definition.*

Definition 2.2 (Deformation of the admissibility bound)**.** *A* deformation *of the admissibility bound is the replacement*

$$A_n \;\leq\; \frac{c_\chi}{\sqrt{\lambda_n}}\,\Phi(\lambda;\,\mathrm{data}_n), \tag{1}$$

where λ is an additional parameter (possibly dimensional) and data_n denotes the BFS-level data at depth n (shell size, spectral weights, Weil block structure). The function Φ is assumed measurable and bounded away from zero and infinity on compact sets.

3 Main Result

Theorem 3.1 (No external dimensional scale, Level 1)**.** *Let the admissibility structure of Foundation M be given with primitive dimensional scale c_χ. Any deformation of the admissibility constraints in the sense of Definition 2.2 that introduces an independent dimensional parameter λ (in the sense of Definition 2.1) breaks at least one of the following:*

(i) BFS-consistent spectral scaling of $\sigma_{\mathrm{pair}}(n)$,

(ii) structural non-injectivity (A2) and the ENI no-go theorem [3],

(iii) bounded admissibility closure (A3–A4).

Proof. We analyse exhaustively all structural forms that the deformation factor Φ can take. Since Φ depends on λ and data_n, there are three cases according to the nature of this dependence.

Case (a): global multiplicative deformation. Suppose $\Phi = f(\lambda)$ is independent of n and of the configuration ω in the fibre.

If λ is an independent dimensional parameter, then $[\lambda]$ is incommensurable with $[c_\chi]$, so $f(\lambda)$ must carry non-trivial dimensions to yield a dimensionally homogeneous right-hand side in (1). But A_n and $c_\chi/\sqrt{\lambda_n}$ are

objects of the same dimensional type by construction of the admissibility bound. Introducing $f(\lambda)$ with $[\, f(\lambda)\,] \neq 1$ produces a right-hand side of different dimensional type, making the bound ill-typed. This violates the closure of A3.

If λ is dimensionless, then $f(\lambda)$ is a dimensionless constant and can be absorbed into a redefinition $c_\chi \mapsto c_\chi\, f(\lambda)^{1/2}$, producing no new scale. The deformation is trivial in this case.

Thus no non-trivial global multiplicative deformation by an independent dimensional parameter is admissible.

Case (b): depth-dependent deformation. Suppose $\Phi = \Phi(\lambda, n)$ depends on the BFS depth n.

The combinatorial growth of BFS shells $|S_n|$ is a property of the Cayley graph $\mathrm{Cay}(G_q, S_q)$ and is not directly modified by the admissibility bound. However, the pair-level projected observable $\sigma_{\mathrm{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ records the effective spectral capacity selected by the admissibility filter at depth n. The paper [1] establishes that $\sigma_{\mathrm{pair}}(n)$ follows a power-law scaling compatible with the Mosco convergence of Q5a [5].

A depth-dependent factor $\Phi(\lambda, n)$ introduces an additional n-dependence in the admissibility selection at each BFS shell. More precisely, the weight assigned to each conjugate Weil block $(c, q-c)$ at depth n is modulated by $\Phi(\lambda, n)$, altering the asymptotic behavior of $\sigma_{\mathrm{pair}}(n)$. Since the power-law scaling of $\sigma_{\mathrm{pair}}(n)$ is determined uniquely by the admissibility structure with scale c_χ and the BFS geometry (established in [1]), any deformation that is not a trivial n-independent rescaling (hence falls back to Case (a)) breaks this scaling. By the Mosco convergence result of Q5a [5], disrupting the power-law scaling of $\sigma_{\mathrm{pair}}(n)$ destroys the existence of the continuum limit, violating closure (i).

Case (c): configuration-dependent deformation. Suppose $\Phi(\lambda, \omega)$ depends on the configuration ω in the projection fibre above a given observable.

A configuration-dependent admissibility filter assigns different selection weights to configurations within the same fibre, i.e., to configurations that share the same observable image under Π. This means that observable-equivalence classes — the projection fibres — are no longer defined by Π alone; the admissibility filter introduces a finer discrimination within each fibre. In other words, $\Phi(\lambda, \omega)$ induces a partial injectivisation: configurations ω, ω' with $\Pi(\omega) = \Pi(\omega')$ but $\Phi(\lambda, \omega) \neq \Phi(\lambda, \omega')$ are no longer genuinely indistinguishable at the admissibility level. This contradicts the structural non-injectivity of A2 and the ENI theorem [3], which requires that distinct underlying configurations sharing the same observable image be assigned identical admissibility weights.

Conclusion. Every deformation by an independent dimensional parameter λ falls into one of Cases (a), (b), or (c), or a combination thereof. In each case the deformation either reduces to a trivial redefinition of c_χ or violates at least one of (i), (ii), (iii). This proves Theorem 3.1. □

4 Corollary: No Additional Field Without Scale

Corollary 4.1 (No additional field without independent dimensional scale)**.** *Under the admissibility structure of Foundation M [4], any additional dynamical field coupled to the projection* Π *and carrying an independent dimensional coupling constant violates Theorem 3.1.*

Proof. Let F be a dynamical field carrying a dimensional coupling constant g_F with $[g_F] \neq [c_\chi]^k$ for any $k \in \mathbb{Q}$. Any dynamical contribution to the admissibility structure must enter the Born–Infeld bound through its energy or action contribution, and therefore modifies the right-hand side of A3. Coupling F to the admissibility structure thus introduces g_F into the right-hand side of A3 as an independent dimensional factor. By Definition 2.1, g_F is an independent dimensional parameter, and by Theorem 3.1 the resulting deformation breaks closure.

Note that fields with *dimensionless* coupling constants (e.g., topological phases, certain geometric angles) are not excluded by the present argument: their coupling factor $\Phi(\lambda, \cdot)$ with $[\lambda] = 1$ falls into Case (a) of the proof and reduces to a trivial rescaling of c_χ. Such fields do not introduce independent dimensional scales and require separate analysis. □

Remark 4.2 (Application to Reissner–Nordström and Schwarzschild–de Sitter)**.** *The electromagnetic charge Q in the Reissner–Nordström metric carries dimensions $[Q]$ independent of $[c_\chi]^k$ for any k. The cosmological constant Λ in the Schwarzschild–de Sitter metric carries dimensions $[\Lambda] = [\text{length}]^{-2}$, likewise independent. By Corollary 4.1, neither admits a consistent coupling to the admissibility structure of Foundation M, which provides the structural basis for the exclusion argument of Remark 3.3 in the companion paper Q6b.*

5 Discussion and Relation to Level 2

Theorem 3.1 establishes that admissibility is rigid with respect to the introduction of new independent dimensional scales. This is a *negative* structural result: it rules out extensions, but does not determine the values of the scales that are already present.

What this result does not say

Theorem 3.1 does not assert that all emergent scales are fully determined by c_χ without residual freedom. Specifically, it leaves open whether derived quantities such as δ_{pair} or β^* — which are of dimensional type compatible with derivation from c_χ via the spectral cascade — are uniquely fixed as functionals of c_χ, or whether they carry unfixed normalisation constants or initial conditions. Resolving this question is the content of Level 2.

Level 2: full functional determination

A Level 2 statement would take the following form:

> *Every quantity of the admissibility cascade is computable from c_χ and the spectral structure of Foundation M, without free parameters or unfixed initial conditions.*

This is a much stronger claim. It requires, in particular, showing that the power-law exponent β^* of the cascade is a determined functional of the BFS geometry and c_χ, not merely a quantity consistent with the scaling established in [1]. Level 2 is left as an open problem.

Derived scales are admissible, not independent

Theorem 3.1 does not exclude the emergence of effective scales from the cascade. Such scales (e.g., mass ratios, spectral thresholds) are admissible precisely because they arise as outputs of the cascade, not as independent inputs. The criterion of Definition 2.1 cleanly separates the two categories: an emergent scale μ is admissible if and only if $[\mu] = [c_\chi]^k$ for some k, or if μ is dimensionless.

Stabilisation of the Gram rigidity classification

The Gram rigidity paper [2] establishes that the admissibility constraint with bound $A_n \leq c_\chi/\sqrt{\lambda_n}$ forces the neutral-generator structure of the projection to be classified by exactly four families of finite subgroups of SU(2): Q_8, Dic_n ($n \geq 3$), $2O$, and $2I$. This classification is derived for the admissibility window defined by c_χ.

Theorem 3.1 provides the missing stability statement: the admissibility window cannot be deformed by an independent dimensional parameter. Consequently, the Gram rigidity classification is not contingent on a particular choice of scale, but is structurally fixed: no deformation of the admissibility bound can produce a different list of compatible groups. The group classification is, in this sense, as rigid as the scale structure itself.

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